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Lecture 5

Chapter 1. The Foundations

- 4. Nested Quantifiers
- 5. Rules of Inference

Previously...

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TABLE 1 Quantifiers.

<i>Statement</i>	<i>When True?</i>	<i>When False?</i>
$\forall x P(x)$	$P(x)$ is true for every x .	There is an x for which $P(x)$ is false.
$\exists x P(x)$	There is an x for which $P(x)$ is true.	$P(x)$ is false for every x .

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TABLE 2 De Morgan's Laws for Quantifiers.

<i>Negation</i>	<i>Equivalent Statement</i>	<i>When Is Negation True?</i>	<i>When False?</i>
$\neg \exists x P(x)$	$\forall x \neg P(x)$	For every x , $P(x)$ is false.	There is an x for which $P(x)$ is true.
$\neg \forall x P(x)$	$\exists x \neg P(x)$	There is an x for which $P(x)$ is false.	$P(x)$ is true for every x .

Nesting of Quantifiers

- Example:

Let the domain of x and y be people.

Let $L(x,y)$ = “ x likes y ” (A statement with 2 free variables – not a proposition)

- Then $\exists y L(x,y)$ = “There is someone whom x likes.” (A statement with 1 free variable x – not a proposition)

- Then $\forall x (\exists y L(x,y))$ =

“Everyone has someone whom they like.”

(A Proposition with 0 free variables.)

Nested Quantifiers

- Nested quantifiers are quantifiers that occur within the scope of other quantifiers.
- The order of the quantifiers is important, unless all the quantifiers are universal quantifiers or all are existential quantifiers.

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TABLE 1 Quantifications of Two Variables.		
<i>Statement</i>	<i>When True?</i>	<i>When False?</i>
$\forall x \forall y P(x, y)$ $\forall y \forall x P(x, y)$	$P(x, y)$ is true for every pair x, y .	There is a pair x, y for which $P(x, y)$ is false.
$\forall x \exists y P(x, y)$	For every x there is a y for which $P(x, y)$ is true.	There is an x such that $P(x, y)$ is false for every y .
$\exists x \forall y P(x, y)$	There is an x for which $P(x, y)$ is true for every y .	For every x there is a y for which $P(x, y)$ is false.
$\exists x \exists y P(x, y)$ $\exists y \exists x P(x, y)$	There is a pair x, y for which $P(x, y)$ is true.	$P(x, y)$ is false for every pair x, y .

Nested Quantifiers

- Let the domain of x and y is $\mathbf{R}$, and $P(x,y): xy = 0$. Find the truth value of the following propositions.

- $\forall x \forall y P(x, y)$ (F)

- $\forall x \exists y P(x, y)$ (T)

- $\exists x \forall y P(x, y)$ (T)

- $\exists x \exists y P(x, y)$ (T)

$\mathbf{R}$: set of real numbers

- $\forall x \exists y P(x,y) \neq \exists y \forall x P(x,y)$

- For every x , there exists y such that $x + y = 0$. (T)

- There exists y such that, for every x , $x + y = 0$. (F)

Nested Quantifiers: Example

- Let the domain = $\{1, 2, 3\}$. Find an expression equivalent to $\forall x \exists y P(x,y)$ where the variables are bound by substitution instead:
 - Expand from inside out or outside in.
 - Outside in:

$$\forall x \exists y P(x,y)$$

$$\equiv \exists y P(1,y) \wedge \exists y P(2,y) \wedge \exists y P(3,y)$$

$$\equiv [P(1,1) \vee P(1,2) \vee P(1,3)] \wedge$$

$$[P(2,1) \vee P(2,2) \vee P(2,3)] \wedge$$

$$[P(3,1) \vee P(3,2) \vee P(3,3)]$$

Quantifier Exercise

- If $R(x,y)$ = “ x relies upon y ,” express the following in unambiguous English when the domain is all people

$\forall x(\exists y R(x,y)) =$ Everyone has *someone* to rely on.

$\exists y(\forall x R(x,y)) =$

There’s a poor overburdened soul whom *everyone* relies upon (including himself)!

$\exists x(\forall y R(x,y)) =$

There’s some needy person who relies upon *everybody* (including himself).

$\forall y(\exists x R(x,y)) =$

Everyone has *someone* who relies upon them.

$\forall x(\forall y R(x,y)) =$

Everyone relies upon *everybody*, (including themselves)!

Negating Nested Quantifiers

- Successively apply the rules for negating statements involving a single quantifier
- Example: Express the negation of the statement $\forall x \exists y (P(x,y) \wedge \exists z R(x,y,z))$ so that all negation symbols immediately precede predicates.

$$\begin{aligned}
 & \neg \forall x \exists y (P(x,y) \wedge \exists z R(x,y,z)) \\
 & \equiv \exists x \neg \exists y (P(x,y) \wedge \exists z R(x,y,z)) \\
 & \equiv \exists x \forall y \neg (P(x,y) \wedge \exists z R(x,y,z)) \\
 & \equiv \exists x \forall y (\neg P(x,y) \vee \neg \exists z R(x,y,z)) \\
 & \equiv \exists x \forall y (\neg P(x,y) \vee \forall z \neg R(x,y,z))
 \end{aligned}$$

Equivalence Laws

- $\forall x \forall y P(x,y) \equiv \forall y \forall x P(x,y)$

$$\exists x \exists y P(x,y) \equiv \exists y \exists x P(x,y)$$

- $\forall x (P(x) \wedge Q(x)) \equiv (\forall x P(x)) \wedge (\forall x Q(x))$

$$\exists x (P(x) \vee Q(x)) \equiv (\exists x P(x)) \vee (\exists x Q(x))$$

- Exercise:

See if you can prove these yourself.

Notational Conventions

- Quantifiers have higher precedence than all logical operators from propositional logic:

$$(\forall x P(x)) \wedge Q(x)$$

- Consecutive quantifiers of the same type can be combined:

$$\forall x \forall y \forall z P(x,y,z) \equiv \forall x,y,z P(x,y,z)$$

$$\text{or even } \forall xyz P(x,y,z)$$

1.5 Rules of Inference

- ***An argument***: a sequence of statements that end with a conclusion
- Some forms of argument (“valid”) never lead from correct statements to an incorrect conclusion. Some other forms of argument (“fallacies”) can lead from true statements to an incorrect conclusion.
- ***A logical argument*** consists of a list of (possibly compound) propositions called premises/hypotheses and a single proposition called the conclusion.
- ***Logical rules of inference***: methods that depend on logic alone for deriving a new statement from a set of other statements. (Templates for constructing valid arguments.)

Valid Arguments (I)

- Example: A logical argument

If I dance all night, then I get tired.

I danced all night.

Therefore I got tired.

- Logical representation of underlying variables:

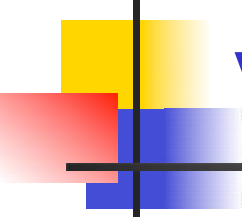
p : I dance all night. q : I get tired.

- Logical analysis of argument:

$p \rightarrow q$ premise 1

p premise 2

$\therefore q$ conclusion

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Valid Arguments (II)

- A form of logical argument is ***valid*** if whenever every premise is true, the conclusion is also true. A form of argument that is not valid is called a ***fallacy***.

Inference Rules: General Form

- An *Inference Rule* is
 - A pattern establishing that if we know that a set of *premise* statements of certain forms are all true, then we can validly deduce that a certain related *conclusion* statement is true.

$\begin{array}{l} \text{premise 1} \\ \text{premise 2} \\ \dots \\ \hline \therefore \text{conclusion} \end{array}$

“ $\therefore$ ” means “therefore”

Inference Rules & Implications

- Each valid logical inference rule corresponds to an implication that is a tautology.

<i>premise 1</i>	Inference rule
<i>premise 2</i>	
<i>...</i>	
$\therefore$ <i>conclusion</i>	

- Corresponding tautology:
 $((\textit{premise 1}) \wedge (\textit{premise 2}) \wedge \square) \rightarrow \textit{conclusion}$

Modus Ponens



$$\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$$

Rule of ***Modus ponens***
(a.k.a. *law of detachment*)

“the mode of affirming”

- $(p \wedge (p \rightarrow q)) \rightarrow q$ is a tautology

p	q	$p \rightarrow q$	$p \wedge (p \rightarrow q)$	$(p \wedge (p \rightarrow q)) \rightarrow q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

- Notice that the first row is the only one where premises are all true

Modus Ponens: Example

If $\left\{ \begin{array}{l} p \rightarrow q : \text{"If it snows today} \\ \text{then we will go skiing"} \\ p : \text{"It is snowing today"} \end{array} \right\}$ assumed TRUE

Then $\therefore q$: "We will go skiing" is TRUE

If $\left\{ \begin{array}{l} p \rightarrow q : \text{"If } n \text{ is divisible by 3} \\ \text{then } n^2 \text{ is divisible by 3"} \\ p : \text{"} n \text{ is divisible by 3"} \end{array} \right\}$ assumed TRUE

Then $\therefore q$: " n^2 is divisible by 3" is TRUE

Modus Tollens

- Rule of *Modus tollens*

$$\begin{array}{l} \neg q \\ p \rightarrow q \\ \hline \therefore \neg p \end{array}$$

“the mode of denying”

- $(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$ is a tautology

- Example

If $\left\{ \begin{array}{l} p \rightarrow q : \text{“If this jewel is really a diamond} \\ \text{then it will scratch glass”} \\ \neg q : \text{“The jewel doesn’t scratch glass”} \end{array} \right\}$ assumed TRUE

Then $\therefore \neg p$: “The jewel is not a diamond” is TRUE

More Inference Rules

$$\frac{p}{\therefore p \vee q}$$

Rule of **Addition**

Tautology: $p \rightarrow (p \vee q)$

$$\frac{p \wedge q}{\therefore p}$$

Rule of **Simplification**

Tautology: $(p \wedge q) \rightarrow p$

$$\frac{p \quad q}{\therefore p \wedge q}$$

Rule of **Conjunction**

Tautology: $[(p) \wedge (q)] \rightarrow p \wedge q$

Examples

- State which rule of inference is the basis of the following arguments:
 - It is below freezing now. Therefore, it is either below freezing or raining now.
 - It is below freezing and raining now. Therefore, it is below freezing now.
- p : It is below freezing now.
 q : It is raining now.
 - $p \rightarrow (p \vee q)$ (rule of addition)
 - $(p \wedge q) \rightarrow p$ (rule of simplification)

Hypothetical Syllogism

$$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$$

Rule of *Hypothetical syllogism*

Tautology:

$$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$$

- Example: State the rule of inference used in the argument:

p
 “If it rains today, then we will not have a q
 barbecue today. If we do not have a barbecue q
 today, then we will have a barbecue tomorrow. r
 Therefore, if it rains today, then we will have a
 barbecue tomorrow.” p
 r

Disjunctive Syllogism

$$\begin{array}{c} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$$

Rule of *Disjunctive syllogism*

Tautology: $[(p \vee q) \wedge (\neg p)] \rightarrow q$

Example

- Ed's wallet is in his back pocket or it is on his desk. $(p \vee q)$ p q
- Ed's wallet is not in his back pocket. $(\neg p)$
- Therefore, Ed's wallet is on his desk. (q)